



OPEN Benchmarking agreement between large language models and published clinical trial conclusions across four artificial intelligence platforms

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Advanced large language models (LLMs) such as ChatGPT, Gemini, Grok3, and Claude offer new possibilities for medical research interpretation and clinical decision support. While these models demonstrate remarkable natural language processing capabilities, their ability to independently reason through clinical trial data and produce conclusions consistent with published trial interpretations remains underexplored. The objective was to evaluate the reliability of LLMs in interpreting numerical and statistical healthcare data. For this study, landmark randomized controlled trials (RCTs) were selected as a standardized domain to minimize bias from poor-quality research designs. Twenty landmark RCTs from the *New England Journal of Medicine* were analyzed in neurosurgical and cardiovascular intervention domains. Four AI platforms were evaluated using a structured prompt covering five domains: evidence interpretation, statistical understanding, clinical relevance, limitation recognition, and practical applicability. Two independent raters scored all AI outputs on a 0–5 scale per domain, and interobserver reliability was assessed. Primary outcomes included concordance with published trial conclusions, accuracy of primary outcome identification, and appropriateness of recommendations. Secondary outcomes included output pattern analysis, recognition of limitations, and handling of confounding factors. ChatGPT demonstrated the highest concordance with published conclusions at 100.0%, followed by Gemini at 84%, Grok3 at 72%, and Claude at 68%. However, these concordance scores should be interpreted cautiously, as the LLMs may have been trained on these published trials, potentially inflating alignment with published conclusions. ChatGPT and Gemini accurately identified limitations and confounding factors, while Grok3 and Claude struggled in these secondary outcomes. Interobserver reliability between raters was good (Cronbach's $\alpha = 0.868$), supporting the consistency of scoring. Certain LLMs, particularly ChatGPT and Gemini, can reliably interpret clinical trial data and align closely with human conclusions, suggesting a potential role in data summarization and evidence synthesis. These findings highlight the importance of selecting AI tools carefully and highlight potential applications in research and clinical workflows.

The advent of large language models (LLMs) such as ChatGPT, Gemini, Grok3, and Claude has ushered in a new era of computational assistance in medicine. These models, capable of processing vast amounts of text, can generate coherent narratives, summarize complex information, and perform reasoning tasks previously limited to humans. Their potential to assist in medical research interpretation and clinical decision support has drawn considerable attention across multiple specialties, including neurosurgery, cardiovascular intervention, and oncology. By analyzing clinical research data, LLMs may streamline evidence synthesis, reduce cognitive burden on clinicians, and facilitate rapid knowledge translation from research into practice^{1–3}. Recent work

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in neurosurgery has already begun to test LLMs for diagnostic support and triage tasks, highlighting both opportunities and limitations in specialty-specific applications⁴.

Despite these capabilities, the extent to which LLMs can independently reason through clinical trial data and arrive at conclusions consistent with human expert interpretation remains unclear. While prior studies have explored AI applications in tasks such as diagnostic support, mortality prediction, and imaging interpretation, there is limited evidence regarding their performance in the structured evaluation of clinical trial data, which often requires integrating statistical findings, clinical context, and trial limitations^{5–7}. Furthermore, there is an ongoing debate regarding the role of AI as an autonomous decision-maker versus a tool to augment human reasoning. It is critical to emphasize that AI should serve primarily as an analytic and data-driven instrument, capable of synthesizing evidence and highlighting relevant trends, while tasks requiring empathy, patient-centered judgment, and nuanced ethical reasoning remain the exclusive domain of human clinicians^{5,6}.

Understanding the concordance between AI-generated interpretations and human expert conclusions is essential for guiding the safe integration of LLMs into research workflows and clinical practice. By establishing the strengths and limitations of these models in interpreting clinical trial data, clinicians and researchers can make informed decisions about leveraging AI as a supplemental tool for evidence-based medicine. The present study evaluates the performance of four LLMs in interpreting landmark clinical trials from neurosurgery and cardiology and benchmarks their concordance with published conclusions across multiple domains^{1,3,6}.

Methods

This study employed a comparative analysis framework involving 20 landmark clinical trials published in the *New England Journal of Medicine* over the past three decades, displayed in Table 1. We examined some of the most highly cited clinical trials in NJEM over the past 20 years. A mixture of medical and surgical trials were selected. Selection criteria included clear primary outcomes, well-defined clinical recommendations, and study designs encompassing randomized controlled trials. One trial was excluded because the results were later invalidated in a subsequent trial. Because landmark NEJM trials were intentionally selected as standardized, high-quality benchmarks, there exists a non-trivial possibility that some or all of these trials were included in the training corpora of the evaluated LLMs.

The following publicly available LLMs were used: ChatGPT o3 (openAI; <https://openai.com>), Grok3 DeeperSearch (xAI; <https://x.ai>), Gemini 2.5 Pro with the deep learning feature (Google; <https://deepmind.google/technologies/gemini>), and Claude 4 (anthropic; <https://www.anthropic.com>). Each model reflects distinct design philosophies, architectures, and training methodologies, allowing for a diverse comparative analysis. ChatGPT and Gemini have been widely recognized for their high performance in natural language reasoning, statistical interpretation, and broad integration into professional workflows, making them strong candidates for clinical research applications. Grok3 was included as a novel entrant emphasizing deeper search and reasoning capacity, offering insight into how emerging platforms compare with more established models. Claude was selected for its focus on safety and alignment, providing a useful counterpoint in evaluating trade-offs between interpretive accuracy and design priorities. The four platforms were selected to capture a representative spectrum of current LLM capabilities, ensuring findings were broadly generalizable while highlighting performance variability in clinical trial interpretation. These model versions represented the most advanced publicly available versions at the time of study conduct in May 2025.

All LLM queries were performed on 5/5/2025 using the model versions specified above. To ensure consistency, all models received identical inputs generated at the same time, and both raters subsequently scored these same archived outputs.

For each trial, the LLMs were prompted to interpret screen captures of tables and figures with their legends from the paper, in the absence of narrative text, with the intention of emphasizing reliance on numerical and statistical outputs for interpretation. Screen captures were provided in standard screenshot format as generated by a screen capture tool. No enhancement or edits were done on the screenshots. While this design sought to constrain interpretation to the provided numerical material, the proprietary nature of LLM training data and internal processes prevented definitive confirmation that outputs were derived exclusively from the supplied screenshots rather than influenced by previously learned information embedded within model parameters.

Each model received the same standardized prompt (Fig. 1), which instructed the AI to analyze the trial results across five domains: evidence interpretation, statistical understanding, clinical relevance, limitation recognition, and practical applicability. “Evidence Interpretation” assessed how accurately the model summarized trial findings and data trends relative to the established trial results. “Statistical Understanding” evaluates the model’s ability to correctly interpret numerical outcomes, confidence intervals, and measures of statistical significance as reported in the trials. “Clinical relevance” examined the model’s capacity to contextualize findings in a manner concordant with the trial’s intended implications for patient care and treatment decisions. “Limitation recognition” evaluates whether the model could identify potential biases, confounders, or methodological constraints as defined by the original study. “Practical applicability” assesses the extent to which the model could translate results into recommendations or guidance that align with the trial’s conclusions and intended clinical application. The complete scoring methodology is illustrated in Fig. 2. The LLMs were explicitly instructed to avoid references to the original paper. All AI outputs were recorded for subsequent comparison.

Two trained analysts evaluated the AI-generated interpretation alongside the original trial manuscript, and each AI output was scored across five categories (evidence, statistics, clinical relevance, limitations, and practical applicability) on a 0–5 scale, where 0 indicated complete absence of capability and 5 indicated perfect alignment. Raters were not blinded to the LLM source; therefore, potential influences of prior perceptions or expectations regarding model performance were not controlled for in the scoring process. Both evaluators scored the same archived LLM outputs to ensure model version consistency. The scoring criteria are further explained in Table 2. Median and interquartile range values were calculated by pooling all individual subsection scores from both

Study	AMA Citation	Intervention	Comparison	Primary Outcome
ACCORD	Action to Control Cardiovascular Risk in Diabetes Study Group, Gerstein HC, Miller ME, et al. Effects of intensive glucose lowering in type 2 diabetes. <i>N Engl J Med.</i> 2008;358(24):2545–2559. doi: https://doi.org/10.1056/NEJMoa0802743	Intensive glucose lowering (HbA1c < 6.0%)	Standard glucose control (HbA1c 7.0–7.9%)	Composite of nonfatal MI, nonfatal stroke, or cardiovascular death
ARISTOTLE	Granger CB, Alexander JH, McMurray JJ, et al. Apixaban versus warfarin in patients with atrial fibrillation. <i>N Engl J Med.</i> 2011;365(11):981–992. doi: https://doi.org/10.1056/NEJMoa1107039	Apixaban 5 mg twice daily	Warfarin (INR 2.0–3.0)	Stroke or systemic embolism
CANTOS	Ridker PM, Everett BM, Thuren T, et al. Antiinflammatory Therapy with Canakinumab for Atherosclerotic Disease. <i>N Engl J Med.</i> 2017;377(12):1119–1131. doi: https://doi.org/10.1056/NEJMoa1707914	Canakinumab (anti-inflammatory therapy)	Placebo	Nonfatal MI, nonfatal stroke, or cardiovascular death
CREST	Brott TG, Hobson RW 2nd, Howard G, et al. Stenting versus endarterectomy for treatment of carotid-artery stenosis. <i>N Engl J Med.</i> 2010;363(1):11–23. doi: https://doi.org/10.1056/NEJMoa0912321	Carotid artery stenting	Carotid endarterectomy	Composite of stroke, MI, or death during periprocedural period and ipsilateral stroke within 4 years
DECRA	Cooper DJ, Rosenfeld JV, Murray L, et al. Decompressive craniectomy in diffuse traumatic brain injury. <i>N Engl J Med.</i> 2011;364(16):1493–1502. doi: https://doi.org/10.1056/NEJMoa1102077	Bifrontotemporoparietal decompressive craniectomy	Standard care	Extended Glasgow Outcome Scale score at 6 months
DESTINY II	Jüttler E, Unterberg A, Woitzik J, et al. Hemispherectomy in older patients with extensive middle-cerebral-artery stroke. <i>N Engl J Med.</i> 2014;370(12):1091–1100. doi: https://doi.org/10.1056/NEJMoa1311367	Hemispherectomy	Standard medical care	Modified Rankin Scale score at 6 months (dichotomized as ≤ 4 vs. > 4)
EMPA-REG	Zinman B, Wanner C, Lachin JM, et al. Empagliflozin, Cardiovascular Outcomes, and Mortality in Type 2 Diabetes. <i>N Engl J Med.</i> 2015;373(22):2117–2128. doi: https://doi.org/10.1056/NEJMoa1504720	Empagliflozin (SGLT2 inhibitor)	Placebo	Composite of death from cardiovascular causes, nonfatal MI, or nonfatal stroke
ENRICH	Pradilla G, Ratcliff JJ, Hall AJ, et al. Trial of Early Minimally Invasive Removal of Intracerebral Hemorrhage. <i>N Engl J Med.</i> 2024;390(14):1277–1289. doi: https://doi.org/10.1056/NEJMoa2308440	Early minimally invasive surgery	Standard medical management	Utility-weighted modified Rankin Scale score at 180 days
IMPROVE-IT	Cannon CP, Blazing MA, Giugliano RP, et al. Ezetimibe Added to Statin Therapy after Acute Coronary Syndromes. <i>N Engl J Med.</i> 2015;372(25):2387–2397. doi: https://doi.org/10.1056/NEJMoa1410489	Ezetimibe plus simvastatin	Simvastatin alone	Composite of cardiovascular death, major coronary event, or nonfatal stroke
INDIGO	Mellinghoff IK, van den Bent MJ, Blumenthal DT, et al. Vorasidenib in IDH1- or IDH2-Mutant Low-Grade Glioma. <i>N Engl J Med.</i> 2023;389(7):589–601. doi: https://doi.org/10.1056/NEJMoa2304194	Vorasidenib (IDH inhibitor)	Placebo	Time to progression or death
JUPITER	Ridker PM, Danielson E, Fonseca FA, et al. Rosuvastatin to prevent vascular events in men and women with elevated C-reactive protein. <i>N Engl J Med.</i> 2008;359(21):2195–2207. doi: https://doi.org/10.1056/NEJMoa0807646	Rosuvastatin 20 mg daily	Placebo	Composite of MI, stroke, hospitalization for unstable angina, arterial revascularization, or cardiovascular death
KEYNOTE	Reck M, Rodríguez-Abreu D, Robinson AG, et al. Pembrolizumab versus Chemotherapy for PD-L1-Positive Non-Small-Cell Lung Cancer. <i>N Engl J Med.</i> 2016;375(19):1823–1833. doi: https://doi.org/10.1056/NEJMoa1606774	Pembrolizumab (immunotherapy)	Platinum-based chemotherapy	Progression-free survival
ODYSSEY	Schwartz GG, Steg PG, Szarek M, et al. Alirocumab and Cardiovascular Outcomes after Acute Coronary Syndrome. <i>N Engl J Med.</i> 2018;379(22):2097–2107. doi: https://doi.org/10.1056/NEJMoa1801174	Alirocumab (PCSK9 inhibitor)	Placebo	Composite of coronary heart disease death, nonfatal MI, fatal or nonfatal ischemic stroke, or unstable angina requiring hospitalization
PARTNER	Leon MB, Smith CR, Mack M, et al. Transcatheter aortic-valve implantation for aortic stenosis in patients who cannot undergo surgery. <i>N Engl J Med.</i> 2010;363(17):1597–1607. doi: https://doi.org/10.1056/NEJMoa1008232	Transcatheter aortic-valve implantation (TAVI)	Standard therapy (medical management)	Composite of death from any cause or repeat hospitalization
PLATO	Wallentin L, Becker RC, Budaj A, et al. Ticagrelor versus clopidogrel in patients with acute coronary syndromes. <i>N Engl J Med.</i> 2009;361(11):1045–1057. doi: https://doi.org/10.1056/NEJMoa0904327	Ticagrelor	Clopidogrel	Composite of death from vascular causes, MI, or stroke
RE-LY	Connolly SJ, Ezekowitz MD, Yusuf S, et al. Dabigatran versus warfarin in patients with atrial fibrillation. <i>N Engl J Med.</i> 2009;361(12):1139–1151. doi: https://doi.org/10.1056/NEJMoa0905561	Dabigatran (110 mg or 150 mg twice daily)	Warfarin	Stroke or systemic embolism
RESCUE	Hutchinson PJ, Kolias AG, Timofeev IS, et al. Trial of Decompressive Craniectomy for Traumatic Intracranial Hypertension. <i>N Engl J Med.</i> 2016;375(12):1119–1130. doi: https://doi.org/10.1056/NEJMoa1605215	Decompressive craniectomy	Standard care	Extended Glasgow Outcome Scale score at 6 months
SPRINT	SPRINT Research Group, Wright JT Jr, Williamson JD, et al. A Randomized Trial of Intensive versus Standard Blood-Pressure Control. <i>N Engl J Med.</i> 2015;373(22):2103–2116. doi: https://doi.org/10.1056/NEJMoa1511939	Intensive blood pressure control (systolic < 120 mm Hg)	Standard control (systolic < 140 mm Hg)	Composite of MI, acute coronary syndrome, stroke, heart failure, or cardiovascular death
STITCH	Jones RH, Velazquez EJ, Michler RE, et al. Coronary bypass surgery with or without surgical ventricular reconstruction. <i>N Engl J Med.</i> 2009;360(17):1705–1717. doi: https://doi.org/10.1056/NEJMoa0900559	CABG with surgical ventricular reconstruction	CABG alone	Death from any cause
SYNAPSE	Skolnick BE, Maas AI, Narayan RK, et al. A clinical trial of progesterone for severe traumatic brain injury. <i>N Engl J Med.</i> 2014;371(26):2467–2476. doi: https://doi.org/10.1056/NEJMoa1411090	Progesterone	Placebo	Glasgow Outcome Scale score at 6 months

Table 1. Clinical Trials provides a list of the 20 landmark clinical trials selected for analysis, along with their corresponding AMA-style citations, interventions, comparisons, and primary outcomes. These trials span neurosurgical and cardiovascular domains and serve as the standardized framework for evaluating LLM performance.

Ready when you are.

With the tables and figures provided, interpret the following: evidence, statistics, clinical relevance, limitations, and practical applicability. Do not reference the original paper this was obtained from



Fig. 1. Prompt This figure illustrates the standardized prompt that was provided to each large language model. The prompt instructed the AI to interpret trial results across five domains while explicitly avoiding references to the original paper. This ensured that all models were evaluated under uniform conditions using only tabular and numerical trial data.

evaluators across the 20 trials, thus preserving rater- and trial-level variability. Inter-rater reliability was assessed by calculating Cronbach's alpha.

Primary outcomes included concordance between AI conclusions and published human conclusions, accuracy in identifying the primary outcomes, and appropriateness of clinical recommendations. Secondary outcomes included assessing reasoning patterns, identifying study limitations, and recognizing potential confounding factors. This study did not involve human participants, animals, or identifiable patient data.

Results

Among the four AI platforms, ChatGPT achieved the highest performance, with a median score of 25 out of 25 (100.0%). Gemini followed with 21 (84%), while Grok3 and Claude scored 18 (72%) and 17 (68%), respectively (Table 3). Across the five evaluation domains, ChatGPT and Gemini consistently demonstrated high concordance in evidence interpretation, statistical understanding, and clinical relevance, while also showing strong recognition of limitations and practical applicability. Grok3 showed moderate performance, excelling in certain domains such as practical applicability but underperforming in evidence interpretation and statistical reasoning. Claude consistently trailed behind the other models across all domains. Interobserver reliability between the two independent raters was good (Cronbach's $\alpha = 0.868$), indicating excellent consistency in LLM response evaluation across all domains. It should be noted that raters were not blinded to the LLM source, meaning that while high inter-rater reliability demonstrated scoring consistency, it did not eliminate the potential for subjective bias introduced by prior expectations regarding model performance.

Despite differences in the individual performances of each LLM, several consistent trends were observed across evaluation domains. ChatGPT, Gemini, Grok3, and Claude all achieved their highest median scores in *clinical relevance* (5/5, 5/5, 4/5, and 4/5, respectively). ChatGPT demonstrated consistently perfect scores (5/5) across all five domains. The lowest median scores varied by model: Gemini scored lowest in *limitation recognition* (4/5), Grok3 scored lowest in both *limitation recognition* and *practical applicability* (3/5 each), and Claude scored lowest in *statistical understanding* (3/5).

Additionally, the evaluation of the responses given by the LLMs was highly consistent for ChatGPT, as evidenced by the narrow interquartile ranges across all domains (Table 3). ChatGPT's scores showed minimal variability, with most domains achieving perfect scores across both raters and all trials. Gemini and Grok3 also demonstrated relatively stable performance, with interquartile ranges typically spanning 0–1 points. In contrast, Claude exhibited the largest variability in its scoring, with interquartile ranges spanning up to 5 points in some domains, reflecting greater inconsistency in its ability to interpret evidence and identify limitations compared with the other models.

Discussion

This study demonstrated that advanced LLMs, particularly ChatGPT and Gemini, can reliably interpret numerical and statistical healthcare data in a manner consistent with expert reasoning, and that high-performing LLMs have the potential to assist with evidence synthesis, knowledge translation, and data-driven decision support in healthcare research workflows¹. Importantly, high concordance in this context should not be interpreted as definitive evidence of independent analytical reasoning, given the possibility of training data contamination. While randomized controlled trials were chosen as the testing framework for this project due to their methodological rigor, the broader purpose was not limited to trial-specific interpretation. Instead, this study highlights the capacity of LLMs to engage in structured reasoning with quantitative healthcare research data, an ability with implications extending beyond the trial setting⁷. Our findings are consistent with other recent evaluations of LLM performance in clinical research, such as ChatGPT's demonstrated concordance in conservative management of idiopathic scoliosis⁸. In addition, LLMs have already been evaluated in specialty

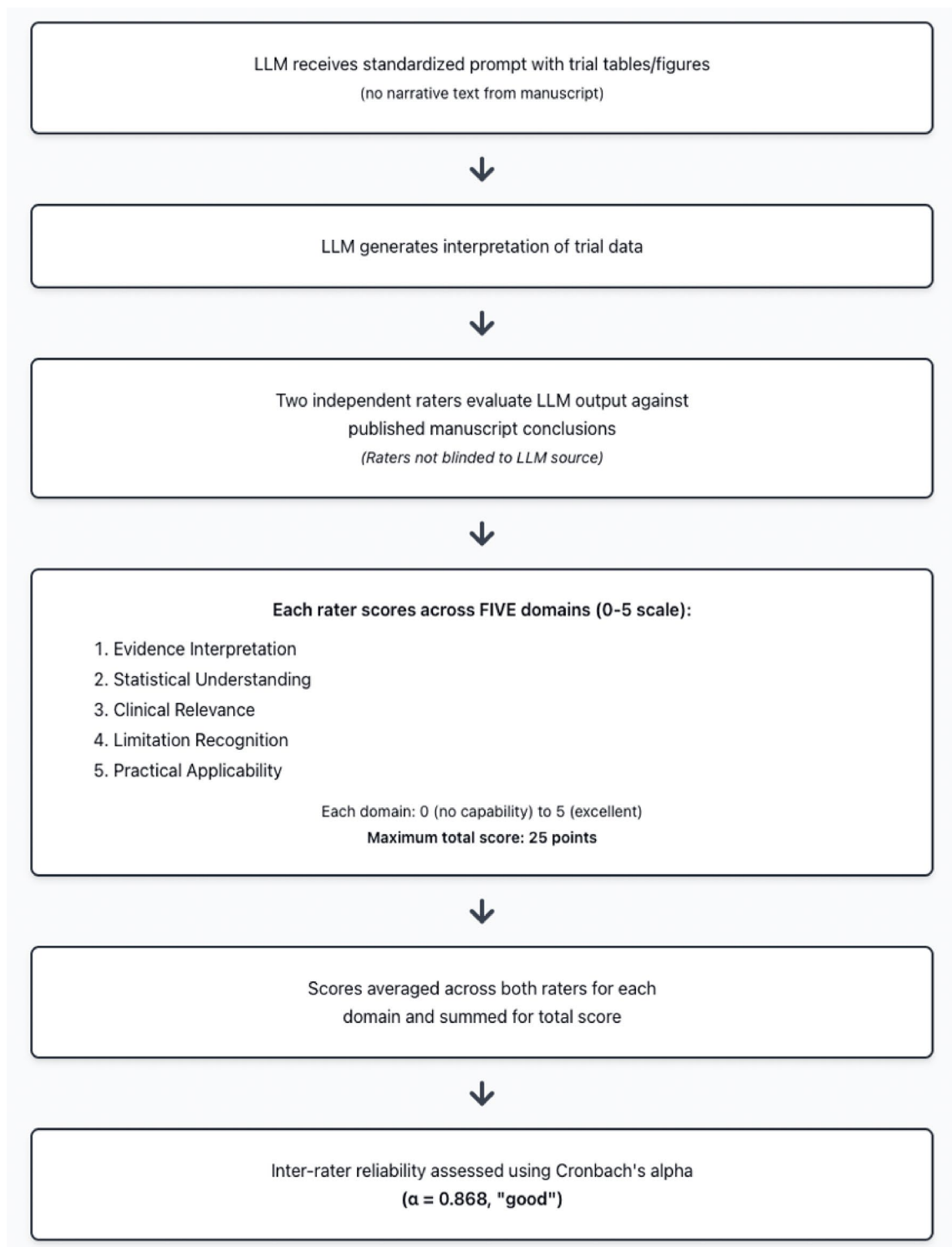


Fig. 2. Scoring Methodology Flowchart. Scoring Scale Definitions (0-5 for each domain):. 0 = Complete absence of capability (no interpretable response). 1 = Minimal capability (mostly inaccurate/irrelevant). 2 = Limited capability (major errors or gaps). 3 = Partial capability (some correct elements, some errors). 4 = Strong capability (mostly accurate with minor gaps). 5 = Excellent capability (comprehensive and accurate)

domains such as neuro-oncology, where they demonstrated potential in generating diagnostic and treatment plans for brain tumor cases, and in neurosurgical triage, where they were assessed for their ability to prioritize patient complaints efficiently^{9,10}.

Performance varied substantially across platforms. ChatGPT showed near-perfect alignment with published conclusions, with Gemini performing well, while Grok3 and Claude's conclusions did not align as well. (Table 3).

Domain	0	1	2	3	4	5
Evidence interpretation	No use of evidence	Minimal, inaccurate	Limited, misses key findings	Partial, some correct evidence	Mostly accurate, minor gaps	Comprehensive and accurate
Statistical understanding	No statistics interpreted	Misinterprets statistics	Major errors, limited clarity	Partial accuracy, some missteps	Mostly correct, minor errors	Complete and accurate
Clinical relevance	No clinical link	Minimal relevance	Weak or indirect link	Partial relevance, some applicability	Mostly relevant with minor gaps	Strong clinical applicability
Limitation recognition	No recognition of limitations	Minimal recognition	Superficial or incomplete	Partial recognition	Recognizes most limitations	Comprehensive and nuanced
Practical applicability	No practical use	Minimal applicability	Limited applicability	Somewhat applicable	Largely applicable	Highly applicable and actionable

Table 2. Scoring Criteria This table summarizes the scoring framework used to evaluate AI-generated trial interpretations. Each domain (evidence interpretation, statistical understanding, clinical relevance, limitation recognition, and practical applicability) was graded on a 0–5 scale, with higher scores reflecting greater accuracy and relevance. The criteria ensured structured, consistent evaluation of LLM performance across multiple dimensions of reasoning and applicability.

Model	Evidence	Statistics	Clinical	Limitations	Practical	Total (Max 25)
ChatGPT o3	5 (5–5)	5 (5–5)	5 (5–5)	5 (5–5)	5 (5–5)	25 (24–25)
Gemini 2.5 Pro	4 (4–4)	4 (4–5)	5 (4–5)	4 (4–4)	4 (4–5)	21 (20–22)
Grok3 DeeperSearch	4 (3–4)	4 (3–4)	4 (4–4)	3 (3–3)	3 (3–4)	18 (17–19)
Claude 4	3 (3–4)	3 (3–3)	4 (3–4)	3 (3–4)	3 (3–4)	17 (14–19)

Table 3. Average ± Standard Deviation of Scores Across LLMs presents the median scores and interquartile ranges for each large language model across the five evaluation domains graded by the two interpreters. ChatGPT achieved the highest overall performance with a median score of 25/25 (100%), followed by Gemini (21/25, 84%), Grok3 (18/25, 72%), and Claude (17/25, 68%), reflecting clear variability in concordance with published conclusions and reliability. The breakdown highlights that ChatGPT demonstrated consistently perfect median scores across all domains, while Gemini scored consistently high, Grok3 performed moderately, and Claude underperformed across the board.

Importantly, two independent raters evaluated all outputs using standardized scoring criteria, and interobserver reliability was strong. This methodological rigor strengthens confidence in the findings, addressing a limitation of earlier exploratory studies where subjective or single-rater evaluation may have introduced bias⁶. These results are also in line with prior studies comparing LLM outputs directly against evidence-based neurosurgeons, which showed overlapping but not identical reasoning approaches. For example, ChatGPT was able to provide treatment recommendations that paralleled expert decisions in many spinal cases, though it occasionally lacked the nuance of guideline-based reasoning¹¹. Given the rapid evolution of LLMs, these specific performance rankings may change with model updates, though the evaluation framework remains applicable to future versions.

The implications of these findings are twofold. First, variability observed across models underscores the need for careful model selection and ongoing oversight, as certain platforms remain unreliable. These observations align with prior work showing LLMs can support diagnostic decision-making. Rajjoub et al. found ChatGPT’s recommendations for lumbar spinal stenosis broadly mirrored expert consensus on conservative-first management but were less reliable when surgical thresholds were considered¹². Similarly, Ahmed et al. reported that ChatGPT’s outputs for degenerative spondylolisthesis were partially concordant with NASS guidelines yet omitted key details, underscoring the risk of incomplete reasoning¹³. Ward et al. further demonstrated ChatGPT’s potential for triaging common spinal complaints, though occasional misses of critical red-flag symptoms highlighted the continued need for human oversight¹⁴.

It is crucial to reinforce that the high scores in the domains of “Clinical relevance” and “Practical Application” reflect textual coherence and alignment with published recommendations based on our scoring rubric. These scores do not represent prospectively validated clinical safety or effectiveness. Future research should be conducted to further explore whether these outputs are safe and effective in a real clinical decision-making scenario. This study demonstrates the variability in outputs from LLMs and provides a domain-based scoring system that future studies can use to evaluate clinical interpretation of AI models in both the research and clinical settings.

Several limitations of this study should be acknowledged. First, although the LLMs were instructed not to reference the original paper from which the data were derived, it is impossible to fully verify that they adhered to this restriction. More fundamentally, the LLMs may have been trained on these published trials, introducing the risk of data contamination that could inflate concordance scores. The true test of independent reasoning would require evaluation on unpublished data or trials published after the models’ knowledge cutoff dates to eliminate training exposure. It should also be noted that ChatGPT’s perfect concordance score (25/25 across all 20 trials) may itself be a potential artifact of training data overlap rather than superior analytical capability,

and should be interpreted with appropriate caution. Additionally, the variability in responses and accuracy supports that the LLMs did not simply access the original article and restate the conclusions; otherwise, it would be expected that all LLMs would have performed similarly in accuracy. Second, the way in which the LLMs were prompted to interpret the findings may have constrained their responses. Without additional context regarding the rationale for conducting the study or the scoring framework, their interpretations were necessarily limited. This lack of background was intentional, as the goal was to minimize contextual scaffolding and assess outputs against structured numerical inputs alone, but it may have restricted the depth and accuracy of their responses. Third, while two researchers independently graded the LLM responses to help mitigate bias, the evaluation process remains inherently subjective and limited by the small number of reviewers. Despite good inter-rater reliability (Cronbach's $\alpha = 0.868$), the scoring involved subjective judgment, and a larger panel of expert raters would strengthen future evaluations. Additionally, reviewers were not blinded to the model source, introducing potential bias. Pre-existing public perceptions of model capabilities, particularly ChatGPT's widespread reputation, may have influenced scoring despite the standardized rubric. The lack of blinding is a significant limitation, particularly given pre-existing public perceptions of model capabilities that could have influenced scoring. Furthermore, if an LLM had generated a novel but correct conclusion that was not addressed by the original authors, our scoring rubric would have penalized it; however, to our knowledge, no such instance occurred. Finally, this study intentionally selected landmark trials with clear, positive findings to establish baseline LLM capabilities. This "best-case scenario" approach may not reflect performance on ambiguous, negative, or methodologically flawed studies, which represents an important area for future research.

Ultimately, while this study employed clinical trial data as a controlled testing environment, the results suggest that advanced LLMs may serve as scalable tools for the broader task of interpreting numerical and statistical healthcare research data, provided their integration is accompanied by rigorous evaluation and human supervision. Future research should evaluate how LLMs perform in their interpretation of more ambiguous and negative clinical trials.

Conclusion

Large language models demonstrated strong performance in this structured evaluation of 20 landmark clinical trials, serving as a benchmark for their ability to interpret numerical and statistical healthcare data. ChatGPT and Gemini achieved the highest concordance with published conclusions, while Grok3 performed moderately and Claude consistently underperformed. It is important to emphasize that this study represents a best-case scenario benchmark: the trials were selected for their clear, positive findings and methodological rigor, and models were evaluated using structured numerical inputs rather than full manuscripts. Because proprietary LLM architectures and training data are not transparent, it was not possible to determine the extent to which outputs were influenced by prior exposure versus independent reasoning. Therefore, the findings reflect baseline or optimal-condition performance rather than guaranteed reasoning independence. These results provide a framework for comparing LLM capabilities and suggest potential utility in data summarization and evidence synthesis, provided human oversight is maintained. Future studies should assess performance on more ambiguous, negative, or methodologically complex trials to evaluate generalizability beyond this benchmark.

Data availability

All data analyzed in this study came from manuscripts published in the New England Journal of Medicine.

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Author contributions

G.M., R.S.S. constructed and supervised the study. W.S., A.S.C., A.S., B.O-G., A.P., L.R.J., and K.J.O.R. collected data from the clinical trials. A.P. and W.S. performed statistical analysis and created figures. W.S. drafted the manuscript. All authors reviewed, edited, and approved the final version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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